MORA-LLM: Enhancing Multiobjective Optimization Recommendation Algorithm by Integrating Large Language Models

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***Abstract*—Multiobjective evolutionary algorithms (MOEAs) have achieved notable success in recommendation systems (RS) by meeting diverse user needs. However, existing MOEAs lack effective methods to coordinate the challenges of cold start, low convergence of multiple objectives and lack of explainable recommendation reasons. Therefore, we propose an enhancing multiobjective optimization recommendation algorithm by integrating large language models (LLMs) (named as multiobjective optimization recommendation algorithm by integrating LLM (MORA-LLM)). MORA-LLM uses the vast knowledge, reasoning ability and natural language generation (NLG) ability of LLMs to compensate for the shortcomings of MOEA-based recommender systems (RSs) in semantic understanding. First, an LLM-enhancing prediction score (LEPS) strategy is proposed to alleviate the cold start problem. LEPS obtains the user embedding vector by vast knowledge of LLM and extracts interaction information of similar users to improve the accuracy of prediction scores. Second, an LLM-enhancing search (LES) strategy is proposed to improve the convergence of the multiobjective. LES strategy combines the reasoning ability of LLM with the competitive idea of competitive swarm optimization to achieve efficient search and balance multiple objectives. Finally, the prediction scores are further corrected based on the LLM output results and MORA-LLM offers recommendation reasons to help users better understand the recommendation results. Experimental results on real-world datasets demonstrate that MORA-LLM significantly outperforms existing algorithms in terms of recommendation accuracy and convergence.**

***Index Terms*—Evolutionary optimization, large language models (LLMs), multiobjective recommendation, recommendation system.**

Received 27 March 2025; revised 30 June 2025 and 20 August 2025; accepted 16 September 2025. Date of publication 19 September 2025; date of current version 1 June 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 61976001, Grant 62502340, and Grant 72471165; in part by the Natural Science Foundation of Anhui Province under Grant 2408085MF152; in part by the Key Projects of the University Excellent Talents Support Plan of Anhui Provincial Department of Education under Grant gxyqZD2021089; and in part by the Natural Science Foundation of Tianjin under Grant 24JCQNJC01560. This article was approved by Associate Editor L. Feng. *(Corresponding author: Lei Zhang.)*

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This article has supplementary downloadable material available at https://doi.org/10.1109/TEVC.2025.3612284, provided by the authors.

Digital Object Identifier 10.1109/TEVC.2025.3612284

I. Introduction

IN TODAY's digital era, the rapid development of the Internet has led to the explosive growth of information and difficulty in classification [1]. To alleviate this problem, recommender systems (RSs) [2] have emerged. Although traditional recommendation algorithms have achieved significant results in RSs, they mainly focus on the accuracy of the recommendations, while ignoring other important nonaccuracy metrics, such as diversity and novelty [3]. Accuracy metric measures the proportion of items that users are interested in, but focusing only on accuracy may lead to over-recommending users' preferred items. To satisfy users' multifaceted needs, RSs need to consider nonaccuracy metrics (e.g., diversity and novelty). However, increasing nonaccuracy metrics often reduces accuracy, leading to conflicts among these metrics [4], [5]. Consequently, RSs are often modeled as multiobjective optimization problems (MOPs) [6], which are typically solved by using multiobjective evolutionary algorithms (MOEAs).

MOEAs are important tools for solving the recommended problems. By simulating the process of natural selection, they optimize multiple objectives, generate Pareto fronts (PFs) and provide diverse and personalized recommendation results for users. For example, Zuo et al. [7] proposed a personalized recommendation framework based on MOEA, which combines multiobjective optimization into the RS to balance the advantages of accuracy and diversity. Wang et al. [8] and Cui et al. [9] further promoted the development of long-tail RS and personalized item evaluation method through multiobjective optimization strategies. Recently, many researchers have explored the combination of multiobjective optimization and other techniques, aiming to further improve the performance of RS. For example, Cai et al. [10], [11] proposed a hybrid RS that combines MOEA with knowledge mining, thereby improving the recommendation quality. Zhao et al. [12] introduced a cooperative-competitive evolutionary algorithm to improve the effect of bidirectional RS by balancing the tradeoff between exploration and exploitation. In addition, the authors [13] proposed an evolutionary algorithm based on community division to solve the challenges in large-scale multiobjective recommendation. Zhang et al. [14], [15], Tian et al. [16] and Ge et al. [17] explored multitask optimization algorithms to further improve the performance of RS in terms of diversity, accuracy and user satisfaction.

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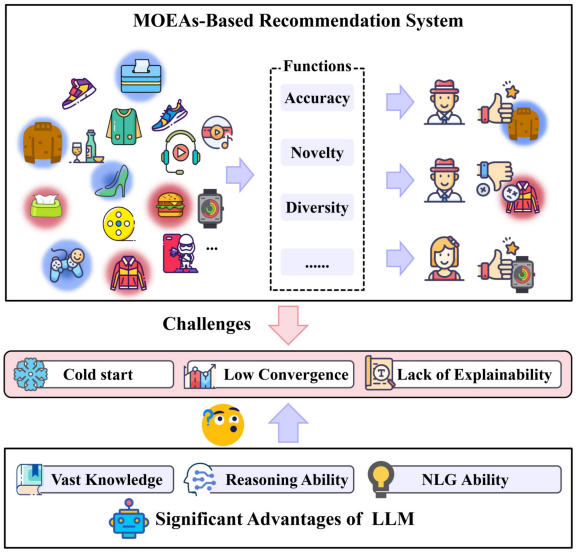


Fig. 1. Motivation of MORA-LLM. The red halo indicates popular items and the black halo indicates unpopular items.

Although MOEAs have achieved remarkable results in RS, there are still some challenges in practical applications. First, the cold start problem becomes more pronounced, especially in the MOEA-based RSs, due to the limited historical interaction data available for new users and the need to learn user preferences from multiple perspectives. Second, the MOEA-based RSs have the potential for further improvement in convergence of multiple objectives and search performance [16]. Finally, the recommendation results lack explainable recommendation reasons [18], which reduces users' trust and understanding of the system.

Based on the above discussion, several challenges in MOEA-based RSs include cold start, low convergence, and lack of explainable recommendation reasons. The advent of large language models (LLMs) presents new opportunities to enhance MOEA-based RSs. This article aims to explore how LLMs can solve the challenges in Fig. 1. First, the difficulty in capturing the preferences of new users is the core problem of the cold start. Based on the vast pretrained knowledge of the LLMs, it can convert users' basic information into embedding vectors [19], thereby identifying potential preferences through similar user data, which significantly improves the accuracy of the RS. Second, based on the reasoning ability of the LLMs, it can generate personalized recommendation lists through designing prompts [20] and context learning [21], thereby enhancing the search performance of multiobjective optimization recommendation algorithms. Finally, LLMs' natural language generation (NLG) ability provides clear understandable explanations for recommendation results. Based on the above considerations, an enhancing multiobjective optimization recommendation algorithm by integrating LLM (MORA-LLM) is proposed. The main contributions are summarized as follows.

1) A MORA-LLM is proposed, which aims to alleviate the cold start problem, improve the convergence and provide explainable recommendation reasons. By leveraging the vast knowledge, reasoning ability and NLG ability of LLMs, MORA-LLM enhances the search performance in multiobjective optimization.
2) An LLM-enhancing prediction score (LEPS) strategy is proposed in MORA-LLM to alleviate the cold start problem. LEPS is based on the vast knowledge of LLM to construct the users' preference embedding vectors, and then extract valuable information from the historical interactions of similar users to compensate for the missing data of new users.
3) An LLM-enhancing search (LES) strategy is proposed to improve the convergence of multiple objectives. First, LES uses LLM as an auxiliary tool to enhance the personalized recommendation performance and search ability of single users. Meanwhile, LES leverages the competitive mechanism of swarm optimization in MOEAs to improve the global convergence. Finally, LES corrects the predicted score based on LLM outputs and provides recommendation reasons to better meet user needs.

The remainder of this article is organized as follows. Section II presents related work on multiobjective recommendation algorithm and the LLM in RSs. Section III describes MORA-LLM. Section IV presents performance experiments for the MORA-LLM, and Section V concludes this article and discusses future research directions.

II. Related Work

This section mainly introduces the multiobjective recommendation problems and the application of multiobjective optimization algorithms in RSs, discusses the introduction of LLM in RSs in recent years.

A. Multiobjective Recommendation Problem

The RS focused on in this article is based on a dataset containing user ratings for items and finding K unrated items for each user to satisfy multiple user requirements. Suppose U is the set of users and I is the set of items. The dataset of user ratings for items can be represented by an interaction matrix R, where the element $R_{i,j}$ represents the rating of the ith user on the jth item (higher rating is better), while many elements of the matrix are unknown (i.e., unrated). Therefore, this recommendation task can be viewed as a multiobjective optimization problem [22], where the objective function represents the requirements of the user

$$\begin{aligned} \max \quad & F(L) = (f_1(L), f_2(L), \ldots, f_M(L)) \\ \text{s.t.} \quad & L = \{L_1, L_2, \ldots, L_{|U|}\} \\ & L_u = \{x_1, x_2, \ldots, x_K\}, x_i \in I \;\; u \in U \end{aligned} \tag{1}$$

where M is the number of objective functions, $|U|$ represents the number of users, K denotes the length of the recommendation list. L_u represents the recommendation list of user u. The solution L represents the recommendation lists of all users and is usually represented in matrix encoding as shown in Fig. 2.

In real-world RSs, just focusing on the accuracy of recommendation results is often insufficient to fully satisfy user

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	x_1	x_2	x_3	x_4	...	x_K
L_1: u_1	5	1	6	10	...	4
L_2: u_2	15	14	7	1	...	9
L_3: u_3	4	6	10	8	...	5
L_4: u_4	1	2	7	11	...	10
... ...	...	...	...	...	...	...
$L_{\lvert U\rvert}$: $u_{\lvert U\rvert}$	14	78	45	11	...	23

Fig. 2. Illustration of matrix encoding.

needs. To improve users' satisfaction, the RS also needs to provide highly innovative and diverse recommendation results. Based on a comprehensive consideration of relevant literature [13], [14], [15], [16], accuracy, novelty and diversity have been selected as the core optimization objectives for the recommendation task in this article.

Accuracy is the most popular metric in a recommendation list and is used to predict the user's rating of items. The higher the user's predicted score for an item represents the more likely the user will like the item. Based the previous work [7], the accuracy function used is described as follows:

$$\text{Accuracy} = \frac{1}{|U| \cdot K} \sum_{u \in U} \sum_{i \in L_u} \hat{R}_{u,i} \tag{2}$$

where $\hat{R}_{u,i}$ is the predicted score of user u for item i. $|U|$ and K are the number of users and recommendation list.

Novelty refers to the proportion of unpopular items in the recommendation list. New items often lack sufficient historical data and are easily overlooked, even though they may better satisfy user needs. Therefore, ensuring higher novelty in recommendations is crucial for improving the performance of RSs. Based on the novelty function in the previous work [16], the novelty function used is described as follows:

$$\text{Novelty} = \frac{1}{|U| \cdot K} \sum_{u \in U} \sum_{i \in L_u} \text{Pop}_{u,i} \tag{3}$$

where $\text{Pop}_{u,i}$ is the popularity of the item i recommended to user u, which is calculated by the inverse of the number of users who have rated the item i in the dataset.

Diversity refers to the differences between items in the recommendation list. In this article, we adopt the topic diversity objective function in the previous work [15], considering that the dataset contains topics of items. The function of topic diversity used is described as follows:

$$\text{Topic Diversity} = \frac{1}{|U|} \sum_{u \in U} H(L_u) \tag{4}$$

$$H(L_u) = \sum_{i \in L_u} \frac{|t_i|}{|z_{L_u}|} \log \frac{|t_i|}{|z_{L_u}|} \cdot \frac{|s_{L_u}|}{N_t} \cdot \sum_{j \in s_{L_u}} q_j \log(q_j) \tag{5}$$

where $|t_i|$ is the number of topics included in item i, $|z_{L_u}|$ represents the number of topics in the recommendation list. The N_t is number of all topic types and the $|s_{L_u}|$ represents the number of topic types in the recommendation list L_u. The q_j is the proportion of topic j in the recommendation list.

B. Multiobjective Optimization in Recommender System

The application of MOEAs in RSs [6], [23] is mainly reflected in balancing accuracy with multiple objectives (such as diversity, novelty and coverage). For example, Zou et al. [24] adopted the multiobjective backtracking search algorithm based on decomposition (MO-BSA). It effectively improves the diversity and accuracy of the RS by decomposing multiple objectives for optimization, especially for personalized recommendations. As the scale of RSs increases, the efficiency of traditional algorithms is low when dealing with large-scale data. To this end, Cai et al. [11] proposed a multiobjective optimization recommendation algorithm based on knowledge mining, which solved the problem of inefficient utilization of knowledge resources in traditional recommendation algorithms. Meanwhile, Cai et al. [10] proposed HMOEA-RS to balance accuracy and diversity. Wei and Wu [25] proposed CC-MOEA, a parallel multiobjective evolutionary algorithm that aims to optimize multiple objectives of RS while improving the efficiency and parallelism of the algorithm. Zhao et al. [12] proposed the CoEA algorithm, which introduced cooperation and competition mechanisms to adjust the contradictions between different roles and improve the quality of the solutions.

To handle larger-scale RSs, Zhang et al. [13] proposed a multiobjective optimization algorithm based on community division, which is specifically aimed at the multiobjective optimization problem and improves the effectiveness of RSs in dealing with large-scale data. Mohammadpour et al. [26] proposed a multiobjective evolutionary method based on NSGA-II to optimize the accuracy and diversity of collaborative filtering (CF) RS. Qiao et al. [27] proposed a feature extraction method in MOEA to optimize multiple recommendation objectives under constraints. Cai et al. [28] designed a knowledge graph-based multiobjective model to optimize accuracy, user satisfaction and novelty in social RS. The authors in [29] proposed a multiobjective refactoring recommendation method based on correlation analysis (MIRROR), which optimizes multiple objectives, such as code quality, redundancy and maintainability. Zhang et al. [14] proposed the multipopulation based evolutionary optimization for RS (MPMORS). This method combines the multipopulation strategy and optimizes for multiple tasks to improve the efficiency and quality of the RS. In addition, Zhang et al. [15] proposed the multitask evolutionary optimization for the multiobjective recommendation algorithm (MOREM), which reduces users and items to construct an auxiliary task for accelerating the search speed of the main task. Meanwhile, Tian et al. [16] proposed an efficient multitask multiobjective evolutionary recommendation algorithm (EMMR), which mainly treats similar users as a task and accelerates search speed through internal and external task information transmission. Finally, considering the computational cost caused by traditional binary encoding methods, Sun et al. [30] designed integer encoding to reduce computation and memory consumption and proposed two archives (convergence-oriented archive CA and diversity-oriented archive DA) to improve recommendation quality.

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In summary, MOEAs are widely used in RSs to balance multiple objectives and improve recommendation performance. However, most of the existing multiobjective optimization recommendation algorithms find it difficult to obtain user preferences when new user interaction data is small. Meanwhile, most multiobjective optimization algorithms lack the transparency of recommendation results.

C. Large Language Models in Recommender System

With the development of the LLM, their potential in natural language processing (NLP) [31] and algorithm design [32] has attracted increasing attention. Zhou et al. [33] first proposed a framework called LLM-assisted design principle transfer (LAPT), which aims to help build a refined search space for new neural architecture search (NAS) tasks by transferring design principles from existing tasks. Subsequently, Wu et al. [34] further explored the fusion mechanism of LLM in evolutionary algorithms (EA) and summarized how LLM and EA mutually enhance each other to achieve better optimization performance. Liu et al. [35] proposed the evolution of heuristics (EoH), which integrates LLMs with evolutionary computation to automatically design high-performance heuristics with a few thousand LLM queries and no domain model training. Wu et al. [36] proposed an algorithm selection model based on LLM (AS-LLM), which combines problem features with algorithm features extracted by LLM to achieve an intelligent algorithm selection mechanism. Yao et al. [37] proposed an LLM-based automated heuristic design framework, which provides more intelligent solutions for complex optimization problems from the perspective of multiobjective optimization.

Due to the vast knowledge and reasoning ability of LLM, LLM can quickly understand user intentions, thus playing a key role in RSs. In recent years, most researchers have focused on using LLMs to enhance or replace traditional RSs [38]. For example, Zhang et al. [39] learned from the interaction mode of dialogue systems, such as ChatGPT, and used the real-time user feedback on recommendation results to generate personalized prompts for sequential recommendation tasks. Yang et al. [40] proposed a novel LLM-based explainable recommendation model, which produces natural language explanations through LLM to improve the transparency of recommendations and user trust. In addition, Yang et al. [41] proposed a sequential recommendation with latent relations based on LLM, which uses LLM to explore the latent relationships and connections between items, thereby improving the modelling ability of RSs for complex user preferences.

Currently, more works are applying LLM to RSs to improve the accuracy and user experience of personalized recommendations [42]. However, integrating LLMs with multiobjective optimization algorithms in RSs is found in literature [43]. The literature [43] primarily focused on the optimization of prompts in conversational RSs, proposed a multiobjective optimization algorithm named LLM-NSGA-II and designed a benchmark problem set named RSBench. LLM-NSGA-II uses the LLM as the core optimization operator and embeds the LLM into the evolutionary process based on NSGA-II. Specifically, the generation of Prompt populations during initialization and the construction of new solutions during descendant generation are both completed by LLM, which makes the entire optimization process more dependent on LLM. Then, the objective values are estimated by randomly selecting samples. However, the output of LLM can be affected by many factors (e.g., input and model state), and relying on the optimization ability of LLM may lead to the output of uncertain results. In contrast, the proposed MORA-LLM algorithm leverages LLMs as auxiliary tools within the recommendation process, rather than integrating them directly into the solution search procedure. Specifically, the LLM is employed in two main components: 1) generating textual embeddings from user information and 2) correcting the predicted scores of candidate recommendation items after the personalized recommendation stage. By utilizing the LLM's extensive knowledge, reasoning capabilities, and NLG strengths, MORA-LLM effectively uncovers users' latent preferences and enhances the quality of recommendation outcomes, thereby improving the overall users' experience.

III. Proposed MORA-LLM

In this section, we first present the general framework of MORA-LLM, followed by a detailed explanation of the proposed LEPS strategy. Subsequently, we describe the proposed LES strategy in detail.

A. Framework of MORA-LLM

Existing multiobjective optimization recommendation algorithms usually rely on users' historical interaction [15] or directly use LLM to generate personalized recommendations [43]. Therefore, the convergence of multiple objectives has a certain potential to be improved and the uncertainty of the recommendation needs to be further reduced. To this end, we propose an enhancing MORA-LLM. The main idea of MORA-LLM is to combine the advantages of LLM and MOEA to optimize multiple objectives of the RS. In the prediction score stage, aiming at alleviating the cold start problem and the scarcity of interaction history, we design an LEPS strategy. LEPS effectively captures the user's interests and preferences based on the user's background information (e.g., user description and historical interaction data) by using LLM to construct the user's embedding vector. In the evolutionary optimization stage, facing the problems of convergence and lack of explainable recommendation reasons, an LES strategy is designed to speed up the search efficiency of the population from all users and the single user.

Based on the above ideas, Fig. 3 shows the main process of the proposed MORA-LLM, which includes two stages: 1) predicted score and 2) evolutionary optimization. The specific steps are described in detail in Algorithm 1. First, in the predicted score stage, the proposed LEPS strategy generates embedding vectors for each user by leveraging the extensive knowledge base of the LLM. LEPS effectively capture the user's interests and preferences and calculates the predicted scores of the items for all users (line 1). Next, in the evolutionary optimization stage, the proposed LES

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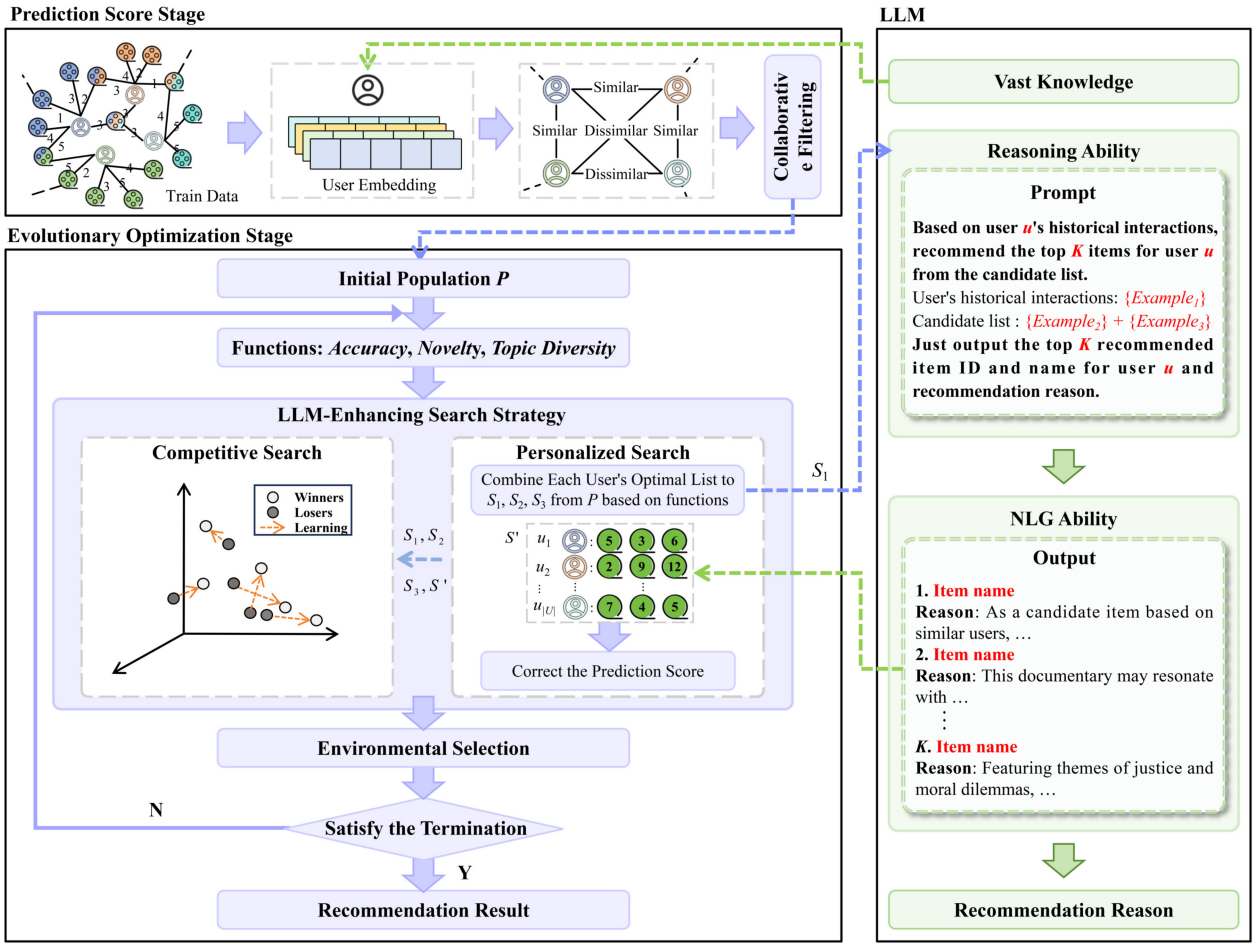


Fig. 3. Framework of MORA-LLM.

Algorithm 1 General Framework of MORA-LLM

Input: *Data*: the dataset; *Popsize*: the population size; *maxGen*: the maximum number of iterations; K: the length of recommendation list;
Output: P: the population;
/*Prediction Score Stage: LEPS Strategy*/
1: $\hat{R} \leftarrow$ Predict the score of all users for all uninteracted items based on LLM;
2: $P \leftarrow$ Items are randomly sampled from the top 100 candidates sorted by predicted scores $\hat{R}$ to generate the initial population;
/*Evolutionary Optimization Stage: LPS Strategy*/
3: **for** $gen = 1$ *to maxGen* **do**
4: $(P_p, Reason) \leftarrow PersonalizedSearch(P, \hat{R}, gen)$;
5: $P_c \leftarrow CompetitiveSearch(P \cup P_p)$;
6: $P \leftarrow$ Select the next generation population from $P \cup P_p \cup P_c$ by the SDE strategy;
7: **end for**
8: **return** P;

strategy mainly comprises personalized search optimization for a single user based on the reasoning ability of LLM (line 4) and competitive search optimization for all users (line 5). Finally, the next generation population is determined by an environmental selection (line 6), selecting excellent solutions to preserve diversity and improve optimization effectiveness. Next, we introduce the LEPS strategy and LES strategy, respectively.

B. LLM-Enhancing Prediction Score Strategy

Most multiobjective EMMRs use ProbS [44] to predict scores, this method is susceptible to the cold start problem, especially when there is little interaction data between users or items. To solve this problem, we propose an LEPS strategy. The LEPS strategy uses the vast knowledge of LLM and combines the user's information (e.g., user description and historical interaction data) to generate the embedding vectors. Specifically, in the process of generating the user embedding vector, LEPS first analyzes the user's basic information (e.g., ID, age, gender, occupation and location), combined with the user's interaction data (e.g., click behavior and rating) on the platform. These pieces of information are input into the LLM (named Mxbai-Embed-Large [19]) and LLM generates personalized user embedding vectors. Because the LLM's embedding uses a fixed text encoder, it consistently generates same embedding vectors for the same input. These embedding vectors contain the basic characteristics and emotional preferences of users. Therefore, LEPS can still infer user interests and preferences in the case of low interaction history.

Specifically, the adopted Mxbai-Embed-Large model generates embedding vectors ($E = \{e_1, \ldots, e_u, \ldots, e_{|U|}\}$) for each

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user and computes similarities between users. In this article, cosine similarity is used to calculate the similarity between user embeddings, which is defined as follows:

$$\text{sim}(u, v) = \frac{e_u \cdot e_v}{||e_u|| \cdot ||e_v||} \tag{6}$$

where e_u and e_v are the embeddings of user u and user v, respectively, $u, v \in U$. $||\cdot||$ denotes the Euclidean norm. The formulation of user-based CF is as follows:

$$\hat{R}_{u,j} = \frac{\sum_{v \in \mathcal{N}_u} \text{sim}(u, v) \cdot R_{v,j}}{\sum_{v \in \mathcal{N}_u} \text{sim}(u, v)} \tag{7}$$

where $\hat{R}_{u,j}$ is the predicted score of user u on item j based on CF. $\mathcal{N}_u$ is the set of users similar to user u. The $R_{v,j}$ is user v rating for item j. $sim(u, v)$ is the similarity between users obtained based on LLM.

Based on the above discussion, MORA-LLM constructs candidate recommendation list and initializes the population for each user based on the predicted score $\hat{R}$ generated in LEPS. Specifically, we select top 100 items as candidate recommendation list for all users based from the $\hat{R}$ in the LEPS strategy. Then, we randomly select K items on the candidate recommendation list and form the initial population $P = \{P_1, P_2, \ldots, P_N\}$, where N is the population size, P_i is the ith solution in the P. Each solution contains the recommendation list results $L = \{L_1, L_2, \ldots, L_{|U|}\}$.

C. LLM-Enhancing Search Strategy

Most multiobjective EMMRs consider all users as a whole by matrix encoding in the optimization process, as shown in Fig. 2. As the number of users and items increases, the search space expands accordingly. This makes it difficult to achieve a balance between objectives, which in turn affects the effectiveness of personalized recommendations. Therefore, a novel LES strategy is proposed to accelerate search and achieve personalized recommendations. LES consists of two main components: 1) personalized search and 2) competitive search. In the personalized search, we introduce how to utilize LLM as an auxiliary tool to enhance search based on user preferences. This includes selecting the optimal recommendation list for each user, constructing prompts and correcting prediction scores. In the competitive search, we introduce how to use the competition idea to speed up the convergence and effectively balance the contradiction between the objectives in the process of recommendation optimization.

1) Personalized Search: In recent years, LLM has shown great potential in NLP and RSs. Traditional recommendation models mainly rely on historical user-item interaction data, which often leads to recommendation results that fail to fully capture user preferences. However, recommendations in the real world often require a deeper understanding of user intentions, preferences and contextual factors that affect decisions. Compared with traditional recommendation algorithms, LLM can more naturally understand user feedback, generate personalized explanations and process complex data, thereby improving user experience. To realize the potential of LLM, we integrate LLM as an auxiliary tool into the multiobjective optimization recommendation system and propose personalized search optimization. It mainly exploits the reasoning and NLG abilities of LLM to achieve personalized search for each user. In Algorithm 2, the personalized search optimization mainly includes three steps: 1) selecting the optimal recommendation list for each user; 2) constructing the prompt; and 3) correcting the prediction score.

Algorithm 2 *PersonalizedSearch*(P, $\hat{R}$, *gen*)

Input: P: the population; $\hat{R}$: the prediction score; *gen*: the current number of iterations; *cycle*: the cycle of using LLM
Output: P_p: the population after personalized search;

$P_p, S_1, S_2, S_3, S', Reason \leftarrow \emptyset$;
for *each user u in P* **do**
  $SP_u \leftarrow$ All solutions in P for user u;
  $S_{1,u}, S_{2,u}, S_{3,u} \leftarrow$ **Select the optimal recommendation list** for user u by the maxinum *Accuracy*, *Novelty* and *Topic Diversity* value from SP_u;
  $S_1 \leftarrow S_1 \cup S_{1,u}$; $S_2 \leftarrow S_2 \cup S_{2,u}$; $S_3 \leftarrow S_3 \cup S_{3,u}$;
  if $gen\%cycle == 0$ **then**
    $(S'_u, Reason) \leftarrow$ **Construct the prompt** for user u and $S_{1,u}$ as shown in Fig. 5 and get the output;
    $S' \leftarrow S' \cup S'_u$;
    $\hat{R}'_u \leftarrow$ **Correct the prediction score** through S and S';
    $\hat{R}_u \leftarrow \hat{R}'_u$
  end if
end for
$P_p \leftarrow S_1 \cup S_2 \cup S_3 \cup S'$;
return P_p, *Reason*;

Selecting the Optimal Recommendation List for Each User: To effectively cope with the token limitations of LLM when processing inputs, it is crucial to avoid a large number of irrelevant or low-quality items in the recommendation list. It helps to improve the efficiency of token utilization. Therefore, selecting the optimal recommendation list for each user is a key step to optimize the recommendation effect. As shown in Fig. 4, the same user recommendation list corresponding to each row is combined from the population P. And then the optimal recommendation list is selected for each user u by the maximum values of *Accuracy*, *Novelty* and *Topic Diversity* and form solutions S_1, S_2, and S_3. Among them, solutions S_1, S_2, and S_3 focus on *Accuracy*, *Novelty* and *Topic Diversity*, respectively.

Constructing the Prompt: When leveraging LLM for personalized recommendation, the right prompt ensures that the LLM understands the recommendation task's purpose and produces results in the desired output format. Specifically, the design of the prompt in this article should guide the model to accomplish two purposes: 1) generating a list of recommended items and 2) providing reasons for each recommendation. Thereby helping users understand why these specific items are recommended. Therefore, when constructing prompts, the output of LLM must fully consider the user's background. First, the prompt needs to include historical interaction data and relevant candidate items to avoid generating false results. In addition, it is crucial for LLM to output recommendation reasons. It can help users understand the logic of recommendations and enhance their acceptance and trust. Finally, it is necessary to clarify the question in the prompt to avoid outputting cluttered

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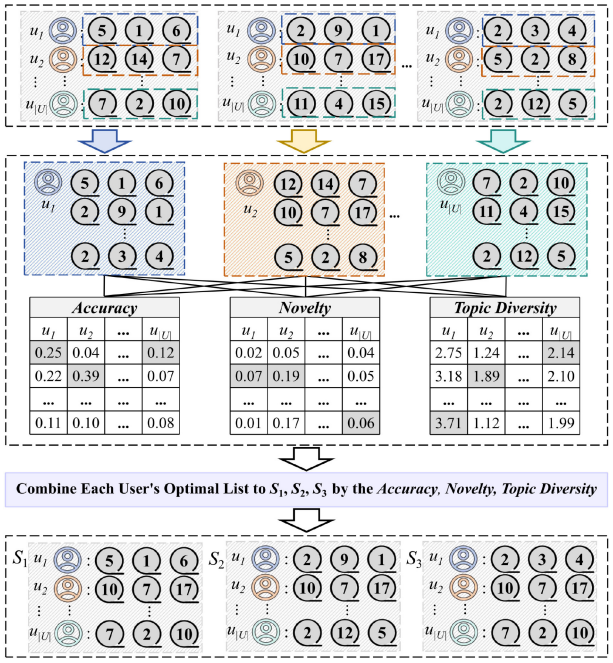


Fig. 4. Example of selecting the optimal recommendation list for each user.

Prompt

Based on user *u*'s historical interactions, the top *K* items are recommended for user *u* from the candidate list.
User's historical interactions: {$Example_1$}
Candidate list : {$Example_2$}+{$Example_3$}
Just output the top *K* recommended item ID and name for user *u* and recommendation reason.

Fig. 5. Prompt template for personalized search.

information. Considering that frequent requests to the LLM can lead to an increase in computing resources, we set the cycle to 20 and use the LLM to provide personalized reasoning for all users at each 20 iterations.

To efficiently build a prompt suitable for LLM, we show an example in Fig. 5. Suppose we need to recommend the top K items for user u. The recommendation results are generated based on this user's historical interactions, user similarity and the current *Accuracy* optimal recommendation solution S_1. The input to the prompt includes multiple replaceable variables ($Example_1$, $Example_2$, $Example_3$) in Fig. 5. It is worth noting that LLM suffers from the token limitation problem and requires careful data screening when designed. Specifically, $Example_1$ is a random selection of 20 items from the user's historical interaction data. $Example_1$ reflects the user's preferences and provide the basis for LLM inference. $Example_2$ is a random selection of 20 items from the candidate recommendation list, which consists of the top 100 items based on prediction scores. It ensures diversity in recommendations while not deviating from users' interests. $Example_3$ is the optimal recommendation list for the user of S_1. $Example_3$ is the items generated by the algorithm and likely to attract the user. To avoid guiding LLM to generate specific results, $Example_2$ and $Example_3$ are inputted together as a candidate list and the items of $Example_2$ and $Example_3$ are not repeated. To constrain the output of the LLM, Fig. 5 includes the instruction "recommend the top K items for user u from the candidate list" in the prompt. After constructing the prompt, we mainly use the reasoning ability and NLG ability of the LLM (Qwen-turbo [45]) to obtain recommendation results and recommendation reasons in line 9 of Algorithm 2. LLM can obtain users' needs and preferences by analyzing their interaction history and recommended projects. Meanwhile, the output of recommendation reasons enhances users' understanding and trust in the recommendation results. It is worth noting that we will verify whether the output of LLM belongs to the candidate list and automatically filter out the recommendation items that are not in the candidate list.

Correcting the Prediction Score: The prediction scores of traditional multiobjective optimization recommendation algorithms are usually fixed, which may cause accurate recommendation results to be deleted in the face of user preferences and environmental changes. In contrast, LLM can extract deeper interest preferences from the user's behavior and interaction history through its strong reasoning ability. Therefore, we dynamically adjust the prediction scores based on the output results of LLM, which can make the recommendation results more in line with the actual needs of users and reduce the recommendation fitting caused by fixed scores. We can effectively integrate the contextual understanding results and historical recommendation results of the LLM through simple weighted averaging, making it more in line with the actual needs of users. The specific formula is as follows:

$$\hat{R}'_{u,i} = \hat{R}_{u,i} + \alpha \frac{\sum\limits_{j \in L_{u,S_1}} \hat{R}_{u,j}}{\left| L_{u,S_1} \right|}, \quad i \in L_{u,S'} / L_{u,S_1} \tag{8}$$

where α is the weight factor to control the degree of correction. $\hat{R}_{u_1}$ is the current prediction score of user u_1 and $\hat{R}'_{u_1}$ is the prediction score modified by (8). The specific process of correcting the prediction score for user u_1 is shown in Fig. 6. Specifically, S_1 is the optimal recommendation list based on *Accuracy* for all users and S' is the output recommendation list for all users based on LLM. The L_{u_1,S_1} is the list of recommended items for user u_1 and the $L_{u_1,S'}$ is the list of recommended items output by LLM for user u_1. Then the average prediction score of items in L_{u_1,S_1} is calculated. Finally, the modified prediction score for item 3 belonging to $L_{u_1,S'}/L_{u_1,S_1}$ is obtained by (8).

2) Competitive Search: Considering the limited exploration ability of recommendation systems in the overall user optimization process, we introduce a competitive concept to optimize the recommendation process. This approach is inspired by competitive swarm optimization (CSO) in MOEAs [46], which has been shown to effectively accelerate convergence in complex optimization problems. The fundamental principle of this competitive concept is that inferior solutions (losers) learn from superior solutions (winners). Winners are selected based on fitness, while losers adjust

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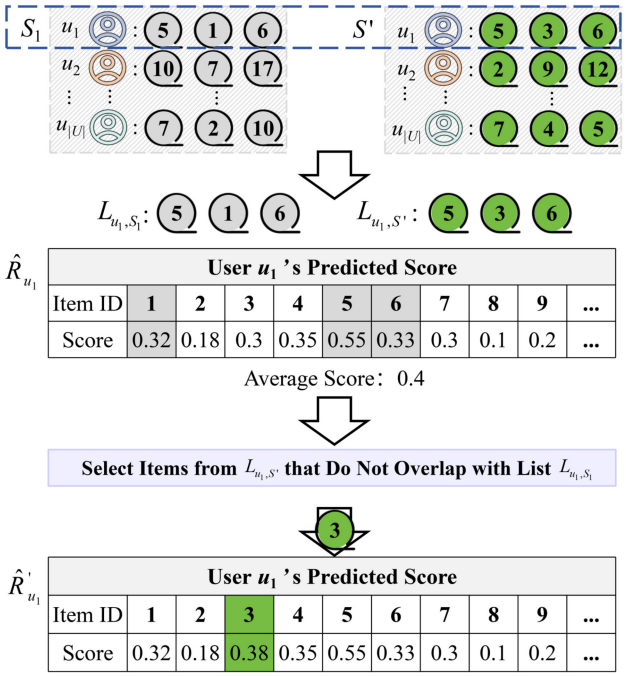


Fig. 6. Example of the correcting the prediction score process.

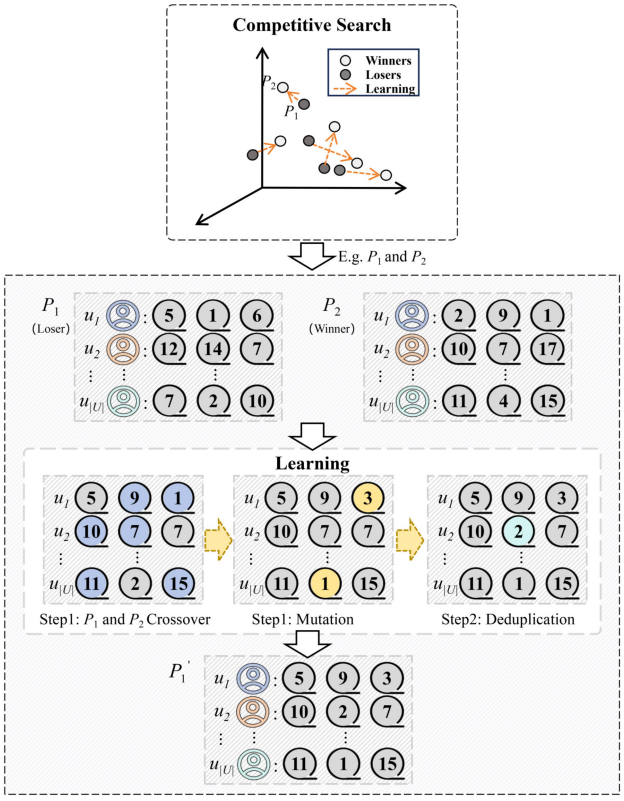


Fig. 7. Example of competitive search process.

their positions. This process accelerates global convergence. We integrate the idea of a competition mechanism into the multiobjective optimization recommendation system to balance the *Accuracy*, *Novelty*, and *Topic Diversity* of the recommendation results.

Algorithm 3 *CompetitiveSearch*($P \cup P_p$)

Input: P: the current population; P_p: the population of personalized search;
Output: P_c: the population after competitive search;

$P_c \leftarrow \emptyset$;
$P \leftarrow P \cup P_p$
$Fit \leftarrow$ Calculate the fitness value of P by the SDE strategy;
for *each solution* P_i *in* P **do**
  $P_j \leftarrow$ Choose any solution randomly from P/P_i;
  $Fit_i, Fit_j \leftarrow$ Get the fitness value of P_i and P_j from Fit;
  if $Fit_i > Fit_j$ **then**
    $Loser \leftarrow P$; $Winner \leftarrow P$;
  else
    $Loser \leftarrow P$; $Winner \leftarrow P$;
  end if
  $P' \leftarrow Loser$ learns from $Winner$;
  $P_c \leftarrow P_c \cup P'$;
end for
return P_c;

To better measure the multiple objectives, MORA-LLM introduces the shift-based density estimation (SDE) [47] strategy to calculate the fitness value of each solution, which better captures the relative importance between objectives. The competitive search is shown in Algorithm 3. First, the fitness value Fit of each solution in the population P ($P \cup P_p$) is calculated by the SDE strategy. Then, a more detailed competitive optimization process is presented in Fig. 7, one solution (e.g., P_2) is randomly selected from P for the current solution (e.g., P_1). P_2 with larger fitness value is selected as the winner and P_1 with smaller fitness value is selected as the loser. To enhance the exploration capability of the optimization process, loser (P_1) learn from winner (P_2), including crossover, mutation and deduplication. It can further search for better solutions. This random selection and competition process is repeated for each solution until optimization is complete.

IV. Experimental Results

This section introduces the experiment settings and then conducts a series of experiments on four real-world datasets, comparing the proposed MORA-LLM with seven representative multiobjective recommendation methods to validate the performance.

A. Experimental Settings

To evaluate the performance of MORA-LLM, we utilized four classic benchmark datasets generated from Movielens and Lastfm. Table I lists the details of the datasets used, including the number of users, items, ratings and topics. For the Movielens-10M, we select 10 000 users as Movielens-10M. The Lastfm dataset contains 1838 users and 11 946 tags. To reduce complexity, we selected 2074 tags with the highest frequency of occurrence after deduplication as the topics.

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TABLE I
STATISTICS OF THESE REAL-WORLD DATASETS

Datasets	#Users	#Items	#Ratings	#Topics
Movielens-100K	943	1,682	100,000	19
Movielens-1M	6,040	3,883	1,000,209	19
Movielens-10M	10,000	10,681	1,435,187	19
Lastfm	1,892	17,632	92,834	2,074

TABLE II
PARAMETER SETTINGS OF COMPARISON ALGORITHMS

Algorithm	N	pc	pm
MOEA-Probs [7]	100	0.9	0.01
PMOEA [9]	100	-	0.01
NewCrossPMOEA [48]	100	-	0.01
ComEA [13]	100	0.9	0.01
MOREM [15]	50	0.9	0.1,0.01
EMMR [16]	10	0.9	0.1
LLM-NSGA-II [43]	10	-	-
MORA-LLM [Ours]	50	0.9	0.1

B. Comparison Algorithms

We compare MORA-LLM[1] with seven representative multiobjective optimization recommendation algorithms, including MOEA-Probs [7], PMOEA [9], NewCrossPMOEA [48], ComEA [13], MOREM [15], EMMR [16], and LLM-NSGA-II [43], as well as four ablation variants (MORA-LLM(-WOLLM), MORA-LLM(-WOM), MORA-LLM(-WOQ), MORA-LLM(-WG)) of MOEA-LLM to validate the effectiveness of different LLM-enhanced components.

1) *MORA-LLM(-WOLLM):* To verify the effectiveness of LLM in MORA-LLM. MORA-LLM (-WOLLM) uses traditional UserCF instead of the LEPS strategy and uses the competitive search in the LES strategy.
2) *MORA-LLM(-WOM):* To verify the effectiveness of LLM in LEPS strategy. MORA-LLM (-WOM) uses traditional UserCF instead of the LEPS strategy.
3) *MORA-LLM(-WOQ):* To verify the effectiveness of LLM in LES strategy. MORA-LLM (-WOQ) only uses the competitive search in the LES strategy.
4) *MORA-LLM(-WG):* To verify the effectiveness of LLM in LES strategy. MORA-LLM (-WG) uses GLM-4-Air [49] model in the LES strategy.

To ensure a fair comparison, all comparison algorithms are set according to the original paper and run independently ten times. The detailed parameter configurations are presented in Table II. The objective functions of all comparison algorithms are (2), (3), and (4). In the MOREM, it is worth noting that the mutation probability pm of the auxiliary task is 0.1 and the mutation probability pm of the original task is 0.01. In LLM-NSGA-II, there is no crossover probability pc and mutation probability pm because the crossover mutation during the generation of offspring is determined by LLM. In particular, EMMR employs LightGCN [50] to estimate the predicted scores for uninteracted items. The LEPS and LES strategies in MORA-LLM use the Mxbai-Embed-Large [19] model and Qwen-Turbo [45] model, respectively. The maximum number of iterations is set to 100. All experimental studies are performed on Intel Core i7-13700K CPU at 3.40 GHz and 64-GB RAM.

C. Evaluation Metrics

In RSs, evaluation metrics are crucial for assessing algorithm performance. *Precision* quantifies the proportion of recommended items that users like

$$\text{Precision} = \frac{1}{|U_t|}\sum_{u\in U_t}\frac{\text{TP}_u}{\text{TP}_u + \text{FP}_u} \tag{9}$$

where U_t is the set of users in the test set. TP_u is the number of items that the user u likes and are recommended. FP_u is the number of items that user u dislike but are recommended. Another important evaluation metric is nonaccuracy, including Novelty and Topic Diversity, as defined by (3) and (4), respectively. In addition, the inverse generation distance (IGD) [51] is commonly used to evaluate the convergence of solutions. In multiobjective recommendation problems, due to the difficulty in obtaining the true PF, we used the maximum values of Precision, Novelty and Topic Diversity to construct a 3-D reference plane. The samll the IGD value, the better the convergence.

D. Effectiveness of the MORA-LLM

1) Precision: In this section, we compared the MORA-LLM with the seven comparison algorithms (MOEA-Probs, PMOEA, NewCrossPMOEA, ComEA, MOREM, EMMR, and LLM-NSGA-II). Table III summarizes the best, mean and standard deviation in *Precision* of all comparison algorithms, respectively, the number of recommendation items K is set to 10 and 20, and the optimal results are bold. The numbers in parentheses represent the ranking of each algorithm on each instance. In Table III, it can be found that among the 16 instances, MORA-LLM gets 12 optimal results and ranks first. NewCrossPMEA and PMEA rank second in the Table III. It is worth noting that NewCrossPMOEA and PMOEA have the same *Precision* on the Movielens-100K and Lastfm, because the upper bound of search results for the top K items in the prediction score is the same.

It can be concluded that our proposed MORA-LLM has better performance in terms of *Precision* than the other seven comparison algorithms. LEPS strategy of MORA-LLM mainly obtains the interactive and descriptive information of similar users through the Mxbai-Embed-Large model without pretraining. This can reduce the decline in the *Precision* of the recommendation results caused by the lack of interaction information for new users. After that, the competitive optimization search of the LES strategy avoids the loss of high precision solutions. The MOEA-Probs, PMOEA, NewCrossPMOEA, ComEA, and MOREM algorithms rely on traditional ProbS [44] in predicting scores. However, ProbS is less effective on sparse datasets. Although EMMR employs

[1]https://anonymous.4open.science/r/MORA-based-LLM-82BE

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TABLE III
Precision FOR ALL COMPARISON ALGORITHMS ON THE FOUR DATASETS

DataSet	K	Qualities	MOEA-Probs	PMOEA	NewCrossPMOEA	ComEA	MOREM	EMMR	LLM-NSGA-II	MORA-LLM
Movielens-100K	10	Best	8.3775e-2 (8)	2.7762e-1 (2)	2.7762e-1 (2)	2.4199e-1 (4)	1.7444e-1 (5)	1.5864e-1 (6)	1.3203e-1 (7)	**2.7932e-1 (1)**
		Mean	7.3457e-2 (8)	2.7762e-1 (2)	2.7762e-1 (2)	2.3881e-1 (4)	1.3906e-1 (6)	1.5549e-1 (5)	1.2463e-1 (7)	**2.7905e-1 (1)**
		Std	4.4570e-3	0.0000e+0	0.0000e+0	3.6403e-3	2.0432e-2	2.1819e-3	4.2244e-3	2.1904e-4
	20	Best	7.3224e-2 (8)	**2.2768e-1 (1)**	**2.2768e-1 (1)**	**2.2768e-1 (1)**	1.0620e-1 (7)	1.5308e-1 (5)	1.3743e-1 (6)	2.2720e-1 (4)
		Mean	6.3828e-2 (8)	**2.2768e-1 (1)**	**2.2768e-1 (1)**	2.2064e-1 (4)	1.0587e-1 (7)	1.5124e-1 (5)	1.3385e-1 (6)	2.2696e-1 (3)
		Std	4.0446e-3	0.0000e+0	0.0000e+0	7.0752e-3	1.1737e-4	1.3444e-3	2.5510e-3	1.7181e-4
Movielens-1M	10	Best	8.5215e-2 (8)	2.3452e-1 (2)	2.3452e-1 (2)	1.8916e-1 (4)	1.5485e-1 (6)	1.7154e-1 (5)	1.1568e-1 (7)	**2.6358e-1 (1)**
		Mean	8.1977e-2 (8)	2.3452e-1 (2)	2.3452e-1 (2)	1.8636e-1 (4)	1.3526e-1 (6)	1.6988e-1 (5)	1.1346e-1 (7)	**2.6357e-1 (1)**
		Std	6.1032e-3	0.0000e+0	0.0000e+0	3.1004e-3	1.0759e-2	1.5833e-3	1.9766e-3	1.4070e-5
	20	Best	7.7525e-2 (8)	1.8860e-1 (2)	1.8860e-1 (2)	1.8860e-1 (2)	9.8046e-2 (7)	1.6757e-1 (5)	1.3092e-1 (6)	**2.1550e-1 (1)**
		Mean	7.4103e-2 (8)	1.8860e-1 (2)	1.8860e-1 (2)	1.8065e-1 (4)	9.8046e-2 (7)	1.6668e-1 (5)	1.2943e-1 (6)	**2.1548e-1 (1)**
		Std	5.5353e-3	0.0000e+0	0.0000e+0	9.2251e-3	0.0000e+0	8.7620e-4	1.1566e-3	1.2463e-5
Movielens-10M	10	Best	7.2930e-2 (8)	2.4467e-1 (2)	2.4467e-1 (2)	1.9661e-1 (4)	1.5806e-1 (5)	1.4421e-1 (6)	1.2876e-1 (7)	**2.6416e-1 (1)**
		Mean	6.3090e-2 (8)	2.4467e-1 (2)	2.4467e-1 (2)	1.8601e-1 (4)	1.2771e-1 (6)	1.4346e-1 (5)	1.0094e-1 (7)	**2.6416e-1 (1)**
		Std	4.7226e-3	0.0000e+0	0.0000e+0	5.8148e-3	2.1504e-2	5.8422e-4	2.8400e-2	0.0000e+0
	20	Best	6.4645e-2 (8)	1.9669e-1 (2)	1.9669e-1 (2)	1.9669e-1 (2)	9.8490e-2 (7)	1.4126e-1 (5)	1.3296e-1 (6)	**2.1056e-1 (1)**
		Mean	5.5493e-2 (8)	1.9669e-1 (2)	1.9669e-1 (2)	1.8491e-1 (4)	9.8490e-2 (7)	1.4088e-1 (5)	1.0320e-1 (6)	**2.1056e-1 (1)**
		Std	4.8409e-3	0.0000e+0	0.0000e+0	5.6678e-3	0.0000e+0	4.4121e-4	3.0760e-2	0.0000e+0
Lastfm	10	Best	3.1660e-2 (8)	**1.5803e-1 (1)**	**1.5803e-1 (1)**	1.1644e-1 (4)	6.3901e-2 (6)	6.5116e-2 (5)	5.5973e-2 (7)	1.5190e-1 (3)
		Mean	2.9075e-2 (8)	**1.5803e-1 (1)**	**1.5803e-1 (1)**	1.1396e-1 (4)	5.7748e-2 (6)	6.2093e-2 (5)	5.3335e-2 (7)	1.5180e-1 (3)
		Std	2.6001e-3	0.0000e+0	0.0000e+0	1.9100e-3	3.0112e-3	1.5637e-3	2.4192e-3	1.7262e-4
	20	Best	2.6797e-2 (8)	1.0139e-1 (2)	1.0139e-1 (2)	1.0139e-1 (2)	4.0539e-2 (7)	6.0703e-2 (5)	4.5587e-2 (6)	**1.0724e-1 (1)**
		Mean	2.4672e-2 (8)	1.0139e-1 (2)	1.0139e-1 (2)	1.0041e-1 (4)	3.9659e-2 (7)	5.9160e-2 (5)	4.3081e-2 (6)	**1.0720e-1 (1)**
		Std	2.3790e-3	0.0000e+0	0.0000e+0	5.4206e-4	3.0921e-4	1.0706e-3	2.1131e-3	3.7477e-5
Average Rank Score			8.0000	1.7500	1.7500	3.4375	6.3750	5.1250	6.5000	**1.5625**

LightGCN, its random population initialization limits early *Precision*. For LLM-NSGA-II, the recommendation is based on the best prompt, but its optimization relies on LLMs. In contrast, MORA-LLM uses LLMs only as auxiliary tools for embedding the generation of LEPS and correcting the prediction score of LES, thereby effectively reducing randomness while leveraging LLMs' semantic capabilities. Therefore, it indirectly verifies that it is meaningful to use LLM as an auxiliary tool in MORA-LLM to enhance the recommendation.

2) Diversity: To verify the diversity of the algorithms in the recommendation results, we summarize the best, mean and standard deviation results of MORA-LLM and seven comparison algorithms on the *Novelty* and *Topic Diversity* in Tables IV and V, respectively. In Tables IV and V, it can be found that MORA-LLM gets 16 optimal results. It is indicated that the LES strategy of MORA-LLM can well balance the contradiction between multiple objectives and can better satisfy the personalized needs of users. It is worth noting that the results of MORA-LLM under the *Novelty* metric are all "1.0000e+0," which is attributed to our specific treatment method when evaluating *Novelty*. To assess the *Novelty* of algorithms, we uniformly set items with a frequency of 0 occurrences to 1. This leads to the fact that the items that have not been evaluated or evaluated once are searched and the final *Novelty* result is 1 in the process of multiobjective optimization search.

3) Convergence: To verify the convergence of the MORA-LLM algorithm in the multiobjective optimization recommendation problem, we summarize the best, mean and standard deviation results of MORA-LLM and seven comparison algorithms on IGD metrics in Tables VI. As can be seen in Tables VI, MORA-LLM consistently achieves the best IGD values among all 16 instances, demonstrating superior convergence toward the true PF. This is attributed to the LES strategy, which integrates the optimal recommendation results for each user, incorporates LLM-based enhancement, and employs competitive search to improve population stability and convergence performance. In contrast, MOEA-Probs, PMOEA, and NewCrossPMOEA use matrix encoding in a large search space, which often loses optimal items and weakens convergence. ComEA, MOREM and EMMR apply group-based search, but the random crossover between parents makes the convergence performance of these algorithms unstable, which limits their effectiveness in complex multiobjective optimization problems. The LLM-NSGA-II algorithm generates diverse prompts through LLMs to flexibly provide personalized recommendations. However, the lack of effective information sharing limits the solution search space, resulting in poor convergence of the population.

For clarity of analysis, Fig. 8 provides a 3-D visualization of the final solution distributions for all algorithms across three objectives (*Precision*, *Novelty* and *Topic Diversity*). As shown in this figure, MORA-LLM effectively maintains high *Precision* while achieving a strong balance between *Novelty* and *Topic Diversity*, demonstrating superior performance in multiobjective optimization. This highlights the effectiveness of the LES strategy in enabling global search, allowing the MORA-LLM to not only optimize for *Precision* but also satisfy diverse recommendation objectives.

4) Explainable Recommendation Reason: The MORA-LLM aims to combine multiobjective optimization and the abilities of LLM to generate personalized recommendations. Different from traditional recommendation systems, the MORA-LLM is not only based on the historical interaction data of users, but also uses the powerful reasoning ability of LLM to generate human-readable recommendation reasons. To show the explainable recommendation reasons provided by MORA-LLM, we take User 1 in the dataset as an example for

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TABLE IV
Novelty FOR ALL COMPARISON ALGORITHMS ON THE FOUR DATASETS

DataSet	K	Qualities	MOEA-Probs	PMOEA	NewCrossPMOEA	ComEA	MOREM	EMMR	LLM-NSGA-II	MORA-LLM
Movielens-100K	10	Best	2.2895e-1 (2)	2.0762e-1 (5)	2.0957e-1 (4)	7.2569e-3 (7)	2.1104e-1 (3)	2.3392e-2 (6)	6.3410e-3 (8)	**1.0000e+0 (1)**
		Mean	2.1392e-1 (2)	2.0541e-1 (4)	2.0335e-1 (5)	7.0863e-3 (7)	2.0600e-1 (3)	2.3011e-2 (6)	6.1280e-3 (8)	**1.0000e+0 (1)**
		Std	1.3213e-2	1.5391e-3	3.3877e-3	1.2238e-4	2.3638e-3	2.9013e-4	1.2775e-4	0.0000e+0
	20	Best	2.1519e-1 (2)	2.0559e-1 (3)	2.0491e-1 (5)	8.1704e-3 (7)	2.0507e-1 (4)	2.1487e-2 (6)	7.5743e-3 (8)	**1.0000e+0 (1)**
		Mean	2.0547e-1 (2)	2.0408e-1 (3)	2.0286e-1 (5)	7.8833e-3 (7)	2.0343e-1 (4)	2.1190e-2 (6)	7.0965e-3 (8)	**1.0000e+0 (1)**
		Std	7.7880e-3	9.8519e-4	1.5318e-3	1.9179e-4	1.4589e-3	1.7874e-4	3.1302e-4	0.0000e+0
Movielens-1M	10	Best	1.3339e-1 (2)	1.3051e-1 (3)	1.3006e-1 (4)	1.1870e-3 (8)	1.2813e-1 (5)	3.0766e-3 (6)	1.3293e-3 (7)	**1.0000e+0 (1)**
		Mean	1.2741e-1 (4)	1.2937e-1 (2)	1.2874e-1 (3)	1.1495e-3 (8)	1.1969e-1 (5)	3.0438e-3 (6)	1.2399e-3 (7)	**1.0000e+0 (1)**
		Std	8.3244e-3	6.8260e-4	6.1779e-4	2.5565e-5	4.4669e-3	1.5141e-5	4.6981e-5	0.0000e+0
	20	Best	1.2841e-1 (4)	1.3065e-1 (2)	1.2965e-1 (3)	1.2383e-3 (8)	1.2114e-1 (5)	2.8784e-3 (6)	1.2688e-3 (7)	**1.0000e+0 (1)**
		Mean	1.2344e-1 (4)	1.2934e-1 (2)	1.2876e-1 (3)	1.2076e-3 (8)	1.1731e-1 (5)	2.8668e-3 (6)	1.2282e-3 (7)	**1.0000e+0 (1)**
		Std	5.5405e-3	6.7551e-4	4.7374e-4	2.0825e-5	4.1687e-3	8.6219e-6	2.0580e-5	0.0000e+0
Movielens-10M	10	Best	2.5185e-1 (2)	2.4975e-1 (4)	2.4876e-1 (5)	1.4336e-3 (7)	2.5004e-1 (3)	2.8387e-3 (6)	1.0415e-3 (8)	**1.0000e+0 (1)**
		Mean	2.4143e-1 (5)	2.4831e-1 (3)	2.4756e-1 (4)	1.3677e-3 (7)	2.4920e-1 (2)	2.8233e-3 (6)	6.4324e-4 (8)	**1.0000e+0 (1)**
		Std	8.9213e-3	1.0748e-3	7.8047e-4	3.2105e-5	8.1663e-4	1.3718e-5	3.4553e-4	0.0000e+0
	20	Best	2.4975e-1 (2)	2.4893e-1 (4)	2.4829e-1 (5)	1.4062e-3 (7)	2.4966e-1 (3)	2.6500e-3 (6)	1.3165e-3 (8)	**1.0000e+0 (1)**
		Mean	2.3775e-1 (5)	2.4816e-1 (3)	2.4713e-1 (4)	1.3851e-3 (7)	2.4857e-1 (2)	2.6450e-3 (6)	7.4891e-4 (8)	**1.0000e+0 (1)**
		Std	9.2545e-3	7.0122e-4	7.9242e-4	9.6360e-6	5.2515e-4	5.0477e-6	4.5737e-4	0.0000e+0
Lastfm	10	Best	7.7629e-1 (2)	7.6959e-1 (3)	7.6666e-1 (5)	1.2670e-1 (6)	7.6945e-1 (4)	1.1510e-1 (7)	4.4972e-2 (8)	**1.0000e+0 (1)**
		Mean	7.7171e-1 (2)	7.6629e-1 (4)	7.6475e-1 (5)	1.1008e-1 (7)	7.6683e-1 (3)	1.1385e-1 (6)	4.3230e-2 (8)	**1.0000e+0 (1)**
		Std	2.9543e-3	2.1365e-3	1.2884e-3	1.1584e-2	1.6382e-3	7.0160e-4	1.7416e-3	0.0000e+0
	20	Best	7.7268e-1 (2)	7.6781e-1 (3)	7.6531e-1 (5)	1.5561e-1 (6)	7.6637e-1 (4)	1.0923e-1 (7)	1.0444e-1 (8)	**1.0000e+0 (1)**
		Mean	7.7061e-1 (2)	7.6597e-1 (3)	7.6473e-1 (5)	1.4501e-1 (6)	7.6570e-1 (4)	1.0855e-1 (7)	1.0279e-1 (8)	**1.0000e+0 (1)**
		Std	1.7723e-3	9.6390e-4	4.5361e-4	7.3445e-3	6.8982e-4	4.8803e-4	1.0193e-3	0.0000e+0
Average Rank Score			2.7500	3.1875	4.3750	7.0625	3.6875	6.1875	7.7500	**1.0000**

TABLE V
Topic Diversity FOR ALL COMPARISON ALGORITHMS ON THE FOUR DATASETS

DataSet	K	Qualities	MOEA-Probs	PMOEA	NewCrossPMOEA	ComEA	MOREM	EMMR	LLM-NSGA-II	MORA-LLM
Movielens-100K	10	Best	3.7235e+0 (5)	3.6207e+0 (7)	3.6824e+0 (6)	4.0262e+0 (2)	3.8055e+0 (4)	3.8863e+0 (3)	3.3145e+0 (8)	**4.9270e+0 (1)**
		Mean	3.6520e+0 (6)	3.5930e+0 (7)	3.6620e+0 (5)	3.9771e+0 (2)	3.7068e+0 (4)	3.8513e+0 (3)	3.2440e+0 (8)	**4.7937e+0 (1)**
		Std	2.6798e-2	1.5029e-2	1.6767e-2	3.4398e-2	6.9287e-2	2.4077e-2	3.1999e-2	1.0118e-1
	20	Best	5.0014e+0 (3)	4.8770e+0 (6)	4.8905e+0 (5)	4.9366e+0 (4)	4.8754e+0 (7)	5.3912e+0 (2)	4.0046e+0 (8)	**8.6804e+0 (1)**
		Mean	4.8309e+0 (5)	4.8158e+0 (6)	4.8443e+0 (4)	4.8550e+0 (3)	4.7855e+0 (7)	5.3545e+0 (2)	3.9582e+0 (8)	**8.3803e+0 (1)**
		Std	7.1580e-2	3.4867e-2	3.6994e-2	3.8047e-2	5.2243e-2	2.0544e-2	3.1042e-2	2.8269e-1
Movielens-1M	10	Best	3.8534e+0 (4)	3.8405e+0 (5)	3.6468e+0 (6)	4.2735e+0 (2)	3.9093e+0 (3)	3.5158e+0 (7)	3.3971e+0 (8)	**4.9496e+0 (1)**
		Mean	3.8343e+0 (3)	3.8292e+0 (4)	3.6346e+0 (6)	4.2076e+0 (2)	3.7811e+0 (5)	3.4974e+0 (7)	3.3717e+0 (8)	**4.8855e+0 (1)**
		Std	1.4239e-2	9.0446e-3	1.3091e-2	3.1146e-2	7.6346e-2	1.2821e-2	1.6656e-2	4.6898e-2
	20	Best	5.1473e+0 (3)	5.1415e+0 (4)	5.1129e+0 (5)	5.0726e+0 (6)	5.1614e+0 (2)	4.9213e+0 (7)	4.3130e+0 (8)	**8.5555e+0 (1)**
		Mean	5.1297e+0 (3)	5.1297e+0 (2)	5.0975e+0 (4)	5.0225e+0 (6)	5.0853e+0 (5)	4.9097e+0 (7)	4.2847e+0 (8)	**8.3774e+0 (1)**
		Std	9.7020e-3	6.4514e-3	1.0104e-2	3.0440e-2	4.4635e-2	1.0180e-2	1.5185e-2	1.0383e-1
Movielens-10M	10	Best	3.8728e+0 (4)	3.8603e+0 (6)	3.8693e+0 (5)	4.0837e+0 (2)	3.8851e+0 (3)	3.7763e+0 (7)	3.4177e+0 (8)	**4.5860e+0 (1)**
		Mean	3.8587e+0 (4)	3.8495e+0 (5)	3.8600e+0 (3)	4.0464e+0 (2)	3.8156e+0 (6)	3.7695e+0 (7)	1.9031e+0 (8)	**4.5564e+0 (1)**
		Std	8.8004e-3	6.6931e-3	6.1726e-3	2.6565e-2	3.4938e-2	6.8834e-3	1.5641e+0	2.3235e-2
	20	Best	4.9461e+0 (2)	4.9453e+0 (3)	4.9376e+0 (4)	4.4587e+0 (7)	4.9203e+0 (5)	4.7345e+0 (6)	3.8963e+0 (8)	**6.7631e+0 (1)**
		Mean	4.9238e+0 (4)	4.9357e+0 (2)	4.9284e+0 (3)	4.4411e+0 (7)	4.9084e+0 (5)	4.7243e+0 (6)	2.1074e+0 (8)	**6.6823e+0 (1)**
		Std	1.2117e-2	5.4328e-3	5.8541e-3	1.0763e-2	8.0192e-3	6.1565e-3	1.8696e+0	6.0486e-2
Lastfm	10	Best	3.3331e+0 (3)	3.3309e+0 (4)	3.3309e+0 (4)	3.6404e+0 (2)	2.7795e+0 (6)	2.1521e+0 (8)	2.4768e+0 (7)	**4.5250e+0 (1)**
		Mean	3.2550e+0 (5)	3.3309e+0 (3)	3.3309e+0 (3)	3.5880e+0 (2)	2.7083e+0 (6)	2.1370e+0 (8)	2.4604e+0 (7)	**4.1849e+0 (1)**
		Std	9.9998e-2	0.0000e+0	0.0000e+0	3.4262e-2	5.6990e-2	8.3786e-3	1.1020e-2	2.2061e-1
	20	Best	4.9174e+0 (3)	4.9164e+0 (4)	4.9164e+0 (4)	4.9453e+0 (2)	4.1014e+0 (6)	3.7228e+0 (7)	3.5823e+0 (8)	**7.0910e+0 (1)**
		Mean	4.7738e+0 (5)	4.9164e+0 (3)	4.9164e+0 (3)	4.9238e+0 (2)	4.0221e+0 (6)	3.7048e+0 (7)	3.5632e+0 (8)	**6.9813e+0 (1)**
		Std	1.6656e-1	0.0000e+0	0.0000e+0	1.6111e-2	6.2857e-2	1.0135e-2	1.6463e-2	9.4339e-2
Average Rank Score			3.8750	4.4375	4.3750	3.3125	5.0000	5.8750	7.8750	**1.0000**

a detailed analysis. The LLM generates clear recommendation reasons that directly reflect user preferences and past behavior in Fig. 9. It is worth noting that we do not constrain the generation of recommendation reasons to preserve the flexibility of the LLM and support more diverse and personalized outputs. For example, if User 1 received a recommendation for "[69]: Forrest Gump," the LLM explain it as "Forrest Gump offers a blend of humor and poignant storytelling about an extraordinary life, aligning perfectly with User 1's appreciation for deep narratives." It reflects the preferences and behavioral patterns of users, making recommendations more tailored to user needs and enhancing personalized experiences. Due to space constraints, the cold start performance comparison of MORA-LLM is provided in Section A of the supplementary material.

E. Further Analysis of MORA-LLM

1) Effectiveness of LLM: To evaluate the effectiveness of LLM in the MORA-LLM, an experimental comparison of MORA-LLM(-WOLLM), MORA-LLM(-WOM), MORA-LLM(-WOQ), MORA-LLM(-WG), and MORA-LLM is

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TABLE VI
IGD FOR ALL COMPARISON ALGORITHMS ON THE FOUR DATASETS

DataSet	K	Qualities	MOEA-Probs	PMOEA	NewCrossPMOEA	ComEA	MOREM	EMMR	LLM-NSGA-II	MORA-LLM
Movielens-100K	10	Best	4.5782e-1 (6)	3.7556e-1 (3)	3.7132e-1 (2)	4.5522e-1 (5)	4.0937e-1 (4)	5.0728e-1 (7)	5.1169e-1 (8)	**2.1670e-1 (1)**
		Mean	4.6372e-1 (6)	3.7619e-1 (3)	3.7214e-1 (2)	4.5677e-1 (5)	4.2836e-1 (4)	5.1062e-1 (7)	5.1456e-1 (8)	**2.2214e-1 (1)**
		Std	4.1100e-3	5.9188e-4	6.8691e-4	7.9802e-4	9.8465e-3	1.4532e-3	1.5986e-3	3.5449e-3
	20	Best	5.1422e-1 (6)	4.2832e-1 (3)	4.2820e-1 (2)	4.8449e-1 (5)	4.8040e-1 (4)	6.0900e-1 (8)	5.3189e-1 (7)	**3.0289e-1 (1)**
		Mean	5.1876e-1 (6)	4.2943e-1 (3)	4.2939e-1 (2)	4.8871e-1 (5)	4.8184e-1 (4)	6.1431e-1 (8)	5.3406e-1 (7)	**3.0867e-1 (1)**
		Std	3.2120e-3	6.6247e-4	1.0058e-3	2.8545e-3	1.0542e-3	3.3158e-3	1.6258e-3	5.7677e-3
Movielens-1M	10	Best	4.8187e-1 (6)	4.0554e-1 (3)	4.0504e-1 (2)	4.8044e-1 (5)	4.4105e-1 (4)	5.1078e-1 (7)	5.2370e-1 (8)	**2.1994e-1 (1)**
		Mean	4.8808e-1 (6)	4.0612e-1 (3)	4.0597e-1 (2)	4.8335e-1 (5)	4.5207e-1 (4)	5.1158e-1 (7)	5.2467e-1 (8)	**2.2709e-1 (1)**
		Std	6.0423e-3	2.6422e-4	4.4845e-4	2.2271e-3	7.4684e-3	4.8417e-4	8.0350e-4	4.9088e-3
	20	Best	5.4231e-1 (6)	4.8218e-1 (3)	4.8022e-1 (2)	5.1308e-1 (5)	4.9446e-1 (4)	5.8948e-1 (8)	5.5360e-1 (7)	**3.2622e-1 (1)**
		Mean	5.4705e-1 (6)	4.8290e-1 (3)	4.8116e-1 (2)	5.1882e-1 (5)	4.9518e-1 (4)	5.9070e-1 (8)	5.5502e-1 (7)	**3.3845e-1 (1)**
		Std	5.2584e-3	4.5166e-4	5.3675e-4	6.8361e-3	4.6420e-4	8.8238e-4	8.1134e-4	7.8847e-3
Movielens-10M	10	Best	4.6506e-1 (5)	3.8474e-1 (3)	3.8042e-1 (2)	4.8204e-1 (6)	4.2007e-1 (4)	5.2450e-1 (8)	5.1895e-1 (7)	**2.7551e-1 (1)**
		Mean	4.7261e-1 (5)	3.8525e-1 (3)	3.8079e-1 (2)	4.8683e-1 (6)	4.3569e-1 (4)	5.2523e-1 (7)	5.6958e-1 (8)	**2.8181e-1 (1)**
		Std	4.8222e-3	4.1985e-4	2.8653e-4	3.3646e-3	1.0501e-2	5.0383e-4	5.2668e-2	3.0374e-3
	20	Best	5.2184e-1 (6)	4.4921e-1 (3)	4.4690e-1 (2)	5.0533e-1 (5)	4.9859e-1 (4)	5.7819e-1 (8)	5.3382e-1 (7)	**3.5103e-1 (1)**
		Mean	5.3052e-1 (6)	4.4982e-1 (3)	4.4785e-1 (2)	5.0986e-1 (5)	4.9927e-1 (4)	5.7863e-1 (7)	5.7983e-1 (8)	**3.5723e-1 (1)**
		Std	4.4055e-3	2.8350e-4	4.6679e-4	2.1956e-3	3.4427e-4	3.0751e-4	4.7732e-2	3.2546e-3
Lastfm	10	Best	3.7270e-1 (5)	2.9792e-1 (3)	2.7783e-1 (2)	4.5585e-1 (6)	3.6365e-1 (4)	4.9517e-1 (7)	5.3166e-1 (8)	**2.5333e-1 (1)**
		Mean	3.7465e-1 (5)	2.9854e-1 (3)	2.7874e-1 (2)	4.5966e-1 (6)	3.6737e-1 (4)	4.9732e-1 (7)	5.3432e-1 (8)	**2.6344e-1 (1)**
		Std	1.4532e-3	6.0546e-4	5.6532e-4	2.8955e-3	3.2276e-3	1.3228e-3	1.8587e-3	6.3863e-3
	20	Best	3.5893e-1 (5)	3.3547e-1 (3)	3.2432e-1 (2)	4.8155e-1 (6)	3.5264e-1 (4)	5.0933e-1 (7)	5.2237e-1 (8)	**2.7709e-1 (1)**
		Mean	3.6141e-1 (5)	3.3673e-1 (3)	3.2606e-1 (2)	4.8756e-1 (6)	3.5416e-1 (4)	5.1086e-1 (7)	5.2419e-1 (8)	**2.8527e-1 (1)**
		Std	1.7495e-3	1.0606e-3	9.9964e-4	4.5067e-3	1.3103e-3	1.0441e-3	1.4632e-3	4.4755e-3
Average Rank Score			5.6250	3.0000	2.0000	5.3750	4.0000	7.3750	7.6250	**1.0000**

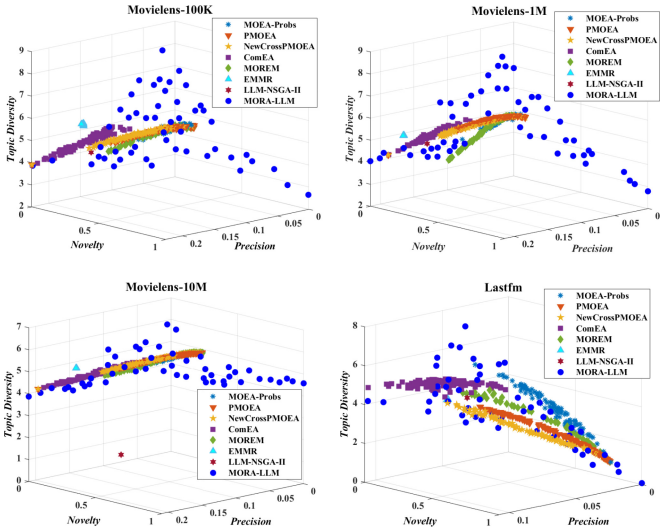


Fig. 8. Final solution set of all comparison algorithms on *Precision*, *Novelty* and *Topic Diversity*.

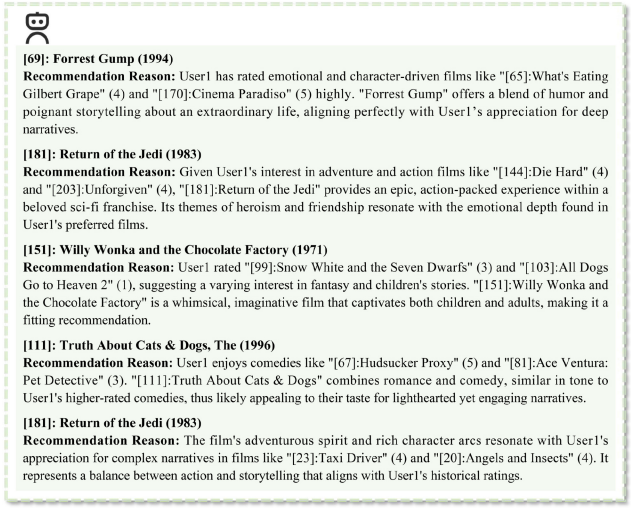


Fig. 9. Example of recommendation result for User 1.

summarized in Fig. 10, where K is set to 20. It can be seen from Fig. 10 that MORA-LLM based on LLM performs better, which verifies the effectiveness of LLM in enhancing recommendation results. The main reason is that personalized search of LES strategy can summarize the optimal recommendation list of each user into a solution, and combine the historical interaction information of users for reasoning and personalized recommendation, which can generate recommended items that are more in line with users' interests. However, there are differences in the impact of different LLMs on recommendation performance. It can be found that MORA-LLM(-WG) based on GLM-4-Air performs more significantly in personalized search performance. This indicates that different models have varying degrees of effectiveness in improving recommendation quality.

To verify the impact of LLM scale on MORA-LLM recommendation performance, we selected Qwen2 [45] series models of different scales (including Qwen2-0.5B, Qwen2-1.5B, Qwen2-7B, and Qwen2-57B) on the Movielens-100K dataset to evaluate their *Precision* and inference time. The final results are summarized in Fig. 11. The experimental results in Fig. 11 indicate that Qwen2-0.5B and Qwen2-1.5B have fewer parameters, their inference time is longer and *Precision* is lower. The main reason for this phenomenon is that Qwen2-0.5B and Qwen2-1.5B run on free service platforms with limited computing resources, resulting in resource sharing and queuing. In contrast, Qwen2-Turbo, Qwen2-7B, and Qwen2-57B are paid services that allocate higher quality

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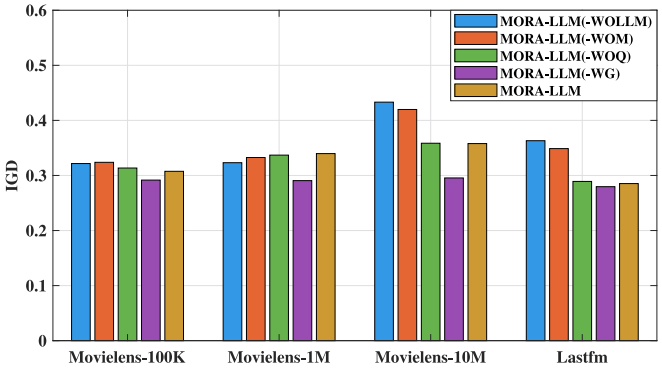


Fig. 10. Mean IGD of MORA-LLM(-WOLLM), MORA-LLM(-WOM), MORA-LLM(-WOQ), MORA-LLM(-WG), and MORA-LLM on the datasets.

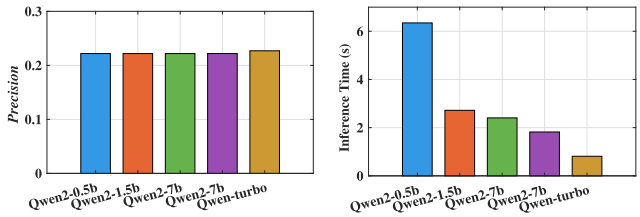


Fig. 11. Mean *Precision* and inference time(s) of each user for LLMs with different model sizes on Movielens-100K.

computing resources, resulting in lower latency and higher throughput. Therefore, we recommend medium-sized models (e.g., Qwen2-Turbo or Qwen2-7B) due to their efficiency and low latency.

2) Runtime: It can be seen that MORA-LLM(-WOQ) without the Qwen-Turbo model has the lowest mean runtime among all algorithms in Fig. 12, while MORA-LLM based on LLM has an overall increase in runtime due to the additional inference and semantic parsing. It is worth noting that the runtime of MORA-LLM with LLM module is still significantly better than traditional multiobjective EMMRs (e.g., PMOEA, NewCrossPMOEA, and ComEA). This indicates that the introduction of LLM has a controllable and reasonable impact on the real-time performance of recommendation systems within an acceptable time cost range.

To assess the efficiency of different LLMs, we summarized the mean inference time of each user on different datasets in Fig.13. For generating the user embedding vector of the LEPS strategy, Mxbai-Embed-Large model consistently achieves the highest *Precision* with stable latency of about 0.6 s, and outperforms the faster and less *Precision* Bert-Base-Uncased, especially on larger datasets. Therefore, we finally chose Mxbai-Embed-Large as the embedding model for the LEPS strategy. In the personalized search stage of LPS, we compare Qwen-Turbo and GLM-4-Air [49]. Qwen-Turbo maintains acceptable real-time performance. In contrast, GLM-4-Air's mean inference time exceeds 2.7 s, which is not conducive to application scenarios with high response speed requirements.

3) Impact of the Weight Factor α: In the LES strategy of the MORA-LLM algorithm, the weight factor α is defined in (8) from the manuscript and plays an important role in the LES. To analyze the effect of the weight factor α in the LES,

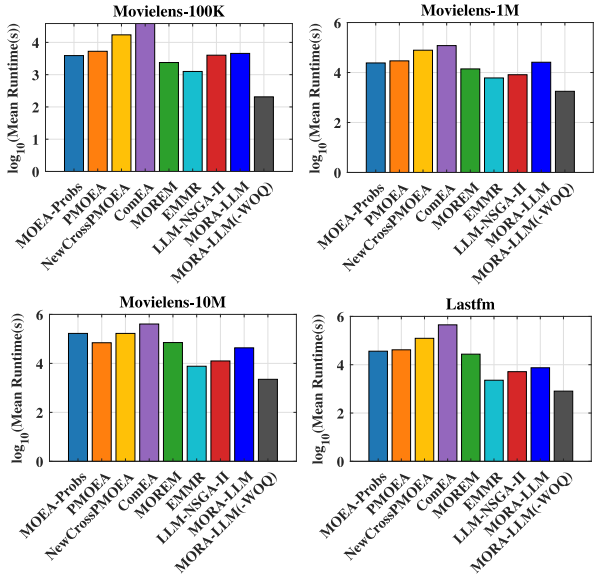


Fig. 12. Mean runtime(s) of all algorithms on the four datasets.

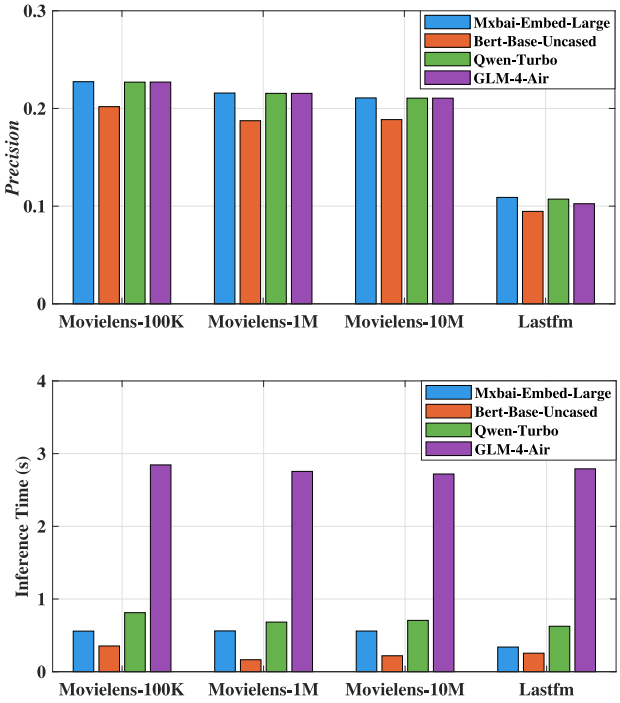


Fig. 13. Mean *Precision* and inference time(s) of each user for LLMs with different model sizes on Movielens-100K.

we adjusted the value of α between 0.1 and 1 at a step of 0.1 and displayed the results in Fig. 14 and K is set to 20. The results indicate that when α is between 0.2 and 0.3, the IGD of MORA-LLM tend to be optimal. Therefore, it is recommended to set α to 0.2 in the MORA-LLM.

V. Conclusion and Future Work

In this article, we proposed an enhanced multiobjective optimization recommendation algorithm by integrating LLMs named MORA-LLM. In MORA-LLM, an LEPS strategy was proposed to alleviate the cold start problem. LEPS obtains

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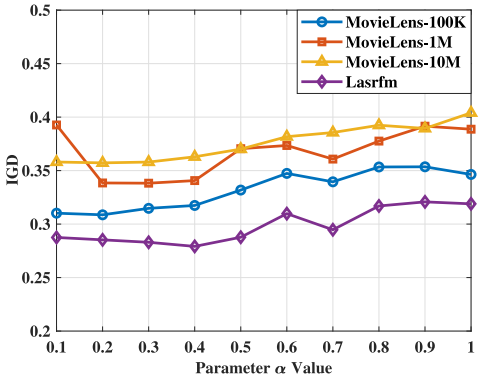


Fig. 14. Experiment on sensitivity parameter α.

the user embedding vector by vast knowledge of LLM and extracts interaction information of similar users to improve the accuracy of prediction scores. Then, an LES strategy was proposed to improve the convergence of multiobjective. LES strategy combines the reasoning ability of LLM with the competitive idea of CSO to achieve efficient search and balance multiple objectives. Finally, the prediction scores were further corrected and recommendation reasons were obtained based on the LLM output results to improve user trust.

Although MORA-LLM performs well in multiobjective recommendation problems, several directions remain for future work. We will explore robustness in extreme cold start scenarios, lightweight model design for deployment, and the use of multimodal data. More importantly, we aim to advance evolutionary computation by investigating LLM-driven operator design and improving the adaptability of EA in dynamic optimization scenarios.

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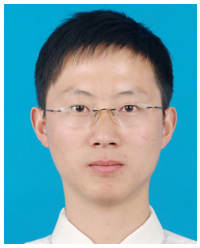
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